

# OUTLIER DETECTION USING IQR

## Student Marks Dataset

### 1. Aim

To identify outliers in a student marks dataset using the Interquartile Range (IQR) method.

### 2. Objective

Calculate Q1 and Q3, derive the IQR, compute the lower and upper IQR fences, and identify marks outside those limits as outliers.

### 3. Dataset

This practical uses a created dataset containing 22 student records.

| Student    | Marks |
|------------|-------|
| Student_01 | 45    |
| Student_02 | 52    |
| Student_03 | 58    |
| Student_04 | 61    |
| Student_05 | 64    |
| Student_06 | 66    |
| Student_07 | 68    |
| Student_08 | 70    |
| Student_09 | 71    |
| Student_10 | 72    |
| Student_11 | 73    |
| Student_12 | 74    |
| Student_13 | 75    |
| Student_14 | 76    |
| Student_15 | 77    |
| Student_16 | 78    |
| Student_17 | 80    |
| Student_18 | 82    |
| Student_19 | 85    |
| Student_20 | 88    |
| Student_21 | 5     |
| Student_22 | 98    |

### 4. IQR Method / Formula

$IQR = Q3 - Q1$ . Lower Bound =  $Q1 - 1.5 \times IQR$ . Upper Bound =  $Q3 + 1.5 \times IQR$ . Values outside these bounds are outliers.

### 5. Calculation

| Measure     | Value |
|-------------|-------|
| Q1          | 64.50 |
| Q3          | 77.75 |
| IQR         | 13.25 |
| Lower Bound | 44.62 |
| Upper Bound | 97.62 |

Outliers are marks below 44.62 or above 97.62.

## 6. Result

2 outlier(s): Student\_21 = 5, Student\_22 = 98

## 7. Output Screenshot – Program Execution

Student Marks Dataset:

| student    | marks |
|------------|-------|
| Student_01 | 45    |
| Student_02 | 52    |
| Student_03 | 58    |
| Student_04 | 61    |
| Student_05 | 64    |
| Student_06 | 66    |
| Student_07 | 68    |
| Student_08 | 70    |
| Student_09 | 71    |
| Student_10 | 72    |
| Student_11 | 73    |
| Student_12 | 74    |
| Student_13 | 75    |
| Student_14 | 76    |
| Student_15 | 77    |
| Student_16 | 78    |
| Student_17 | 80    |
| Student_18 | 82    |
| Student_19 | 85    |
| Student_20 | 88    |
| Student_21 | 5     |
| Student_22 | 98    |

IQR OUTLIER DETECTION

Q1 (25th percentile): 64.50

Q3 (75th percentile): 77.75

IQR: 13.25

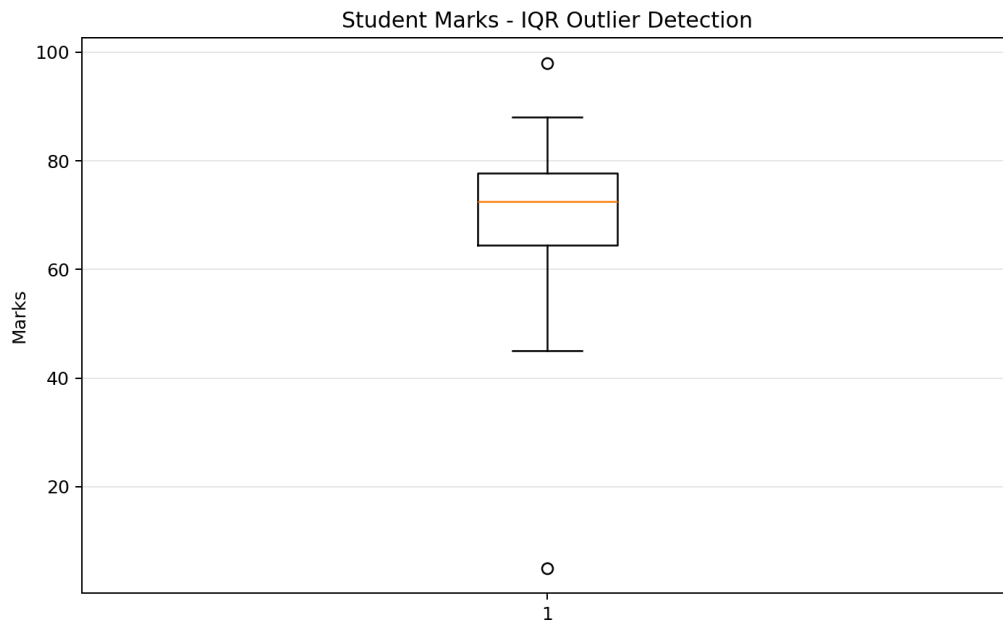
Lower Bound: 44.62

Upper Bound: 97.62

Outlier Records:

| student    | marks |
|------------|-------|
| Student_21 | 5     |
| Student_22 | 98    |

## 8. Output Screenshot – Boxplot



## 9. Full Working Python Code

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("student_marks.csv")
Q1 = df["marks"].quantile(0.25)
Q3 = df["marks"].quantile(0.75)
IQR = Q3 - Q1
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR
outliers = df[(df["marks"] < lower_bound) | (df["marks"] > upper_bound)]

print("Student Marks Dataset:")
print(df.to_string(index=False))
print("\nIQR OUTLIER DETECTION")
print(f"Q1 (25th percentile): {Q1:.2f}")
print(f"Q3 (75th percentile): {Q3:.2f}")
print(f"IQR: {IQR:.2f}")
print(f"Lower Bound: {lower_bound:.2f}")
print(f"Upper Bound: {upper_bound:.2f}")
print("\nOutlier Records:")
print(outliers.to_string(index=False))

plt.figure(figsize=(8,5))
plt.boxplot(df["marks"])
plt.title("Student Marks - IQR Outlier Detection")
plt.ylabel("Marks")
plt.grid(axis="y", alpha=0.3)
plt.tight_layout()
plt.savefig("screenshots/iqr_boxplot.png", dpi=180)
plt.show()
```

## 10. Conclusion

The IQR method detected 2 outlier(s). The outlier marks are 5, 98.